3 An object is suspended from a spring that is attached to a fixed point as shown in Fig. 3.1.

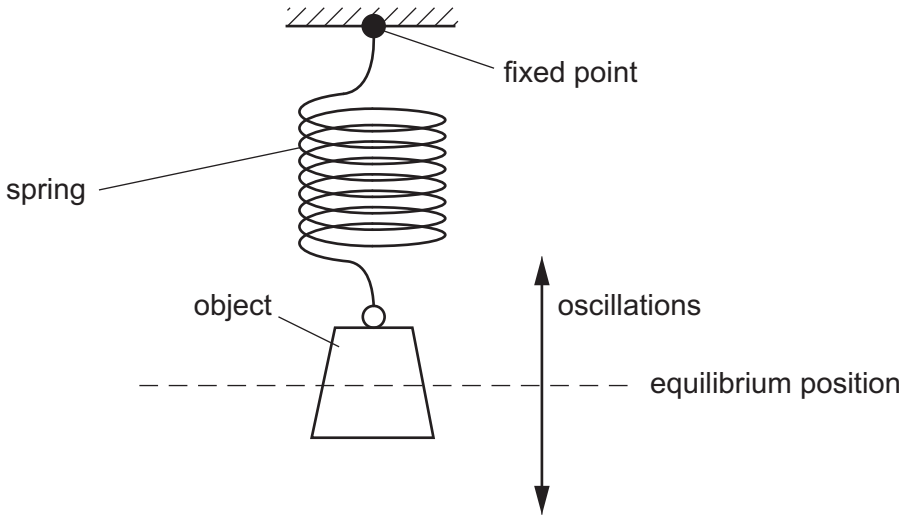


Fig. 3.1

The object oscillates vertically with simple harmonic motion about its equilibrium position.

(a) State the defining equation for simple harmonic motion. Identify the meaning of each of the symbols used to represent physical quantities.

.....

.....

..... [2]

(b) The variation with displacement x from the equilibrium position of the velocity v of the object is shown in Fig. 3.2.

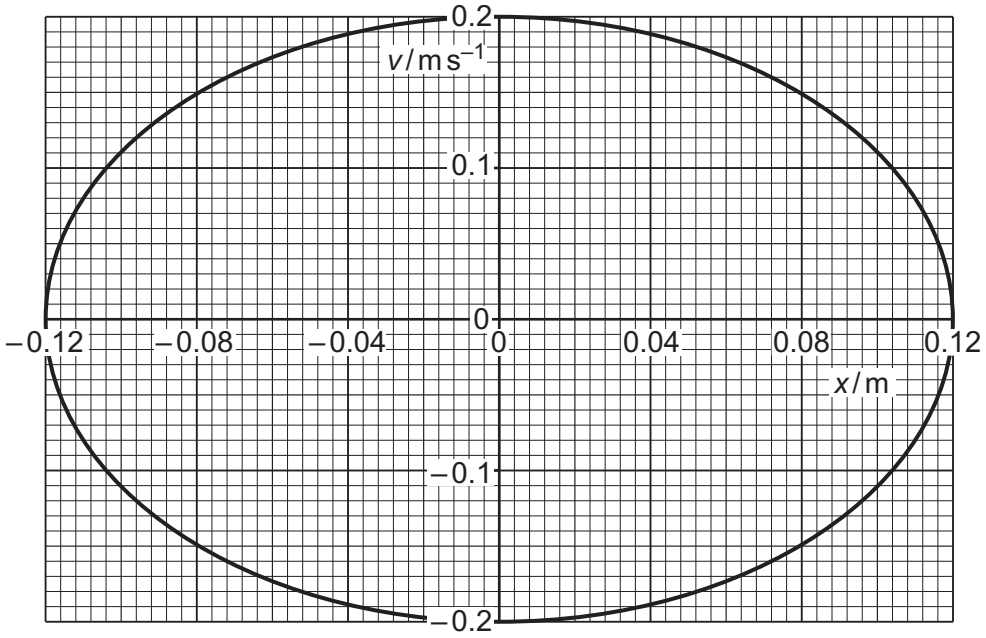


Fig. 3.2

The variation with x of the potential energy E_P of the oscillations of the object is shown in Fig. 3.3.

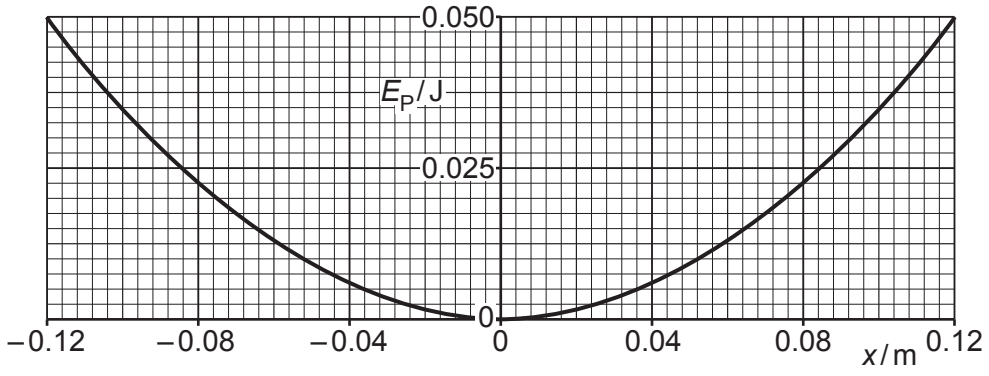


Fig. 3.3

Use Fig. 3.2 and Fig. 3.3 to:

(i) determine the amplitude x_0 of the oscillations

x_0 = ... m [1]

(ii) show that the angular frequency of the oscillations is 1.7 rad s^{-1}

[2]

(iii) determine the mass M of the object.

M = ... kg [2]

(c) The oscillations of the object are now lightly damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) Assume that the damping does not change the angular frequency of the oscillations.

On Fig. 3.2, sketch the variation with x of v when the amplitude of the oscillations is 0.060 m. [2]

[Total: 11]